

Cortex-M Online Programmer

User Manual

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1 Introduction

1.1 Overview

CM ISP (Cortex-M In-System Programmer) is an online programmer software for Cortex-M series MCU of Xiaohua Semiconductor (XHSC), supporting all Cortex-M series MCU products of Xiaohua Semiconductor. This article will introduce the usage of the online programmer software (xhsc.exe) and programming precautions.

This article is applicable to the online programmer software version number V2.02 and above.

1.2 Connection Overview

When using CM ISP programmer software (xhsc.exe), the connection between the serial port module and the target MCU is shown in Figure 1-1.

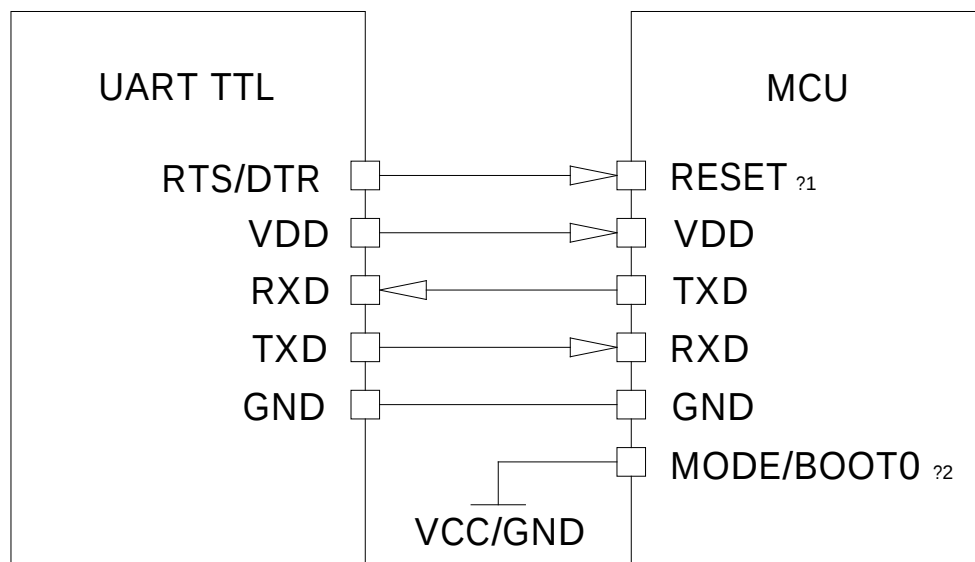


Figure 1-1 Connection between the serial port module and the target MCU

Connect the target chip to the serial port module, and then connect the serial port module to the PC.

Notes:

- Some series do not have MODE/BOOT0 pins. When serial programming, the RTS or DTR of the serial port module needs to be connected to the RESET pin of the target MCU.
- The connection method of the MODE/BOOT0 pin of different chip models may be different. Please refer to the programmer software interface prompt for details.

1.3 Software Operation Overview

The operating environment of the online programmer software is shown in Table 1-1.

Table 1-1 Operating environment of the programmer software

Operating system	Windows 7, Windows 8, Windows 10
Framework version	Framework 4.0 or above

Running the software requires that Microsoft.NET Framework v4.0 or above has been installed on the computer. Please confirm whether Framework 4.0 exists in the computer system path "C:\Windows\Microsoft.NET\Framework(64)", as shown in Figure 1-2.

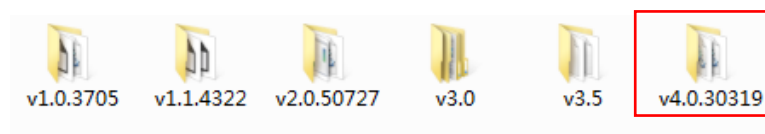


Figure 1-2 Framework 4.0

If the operating system is not installed, please select the corresponding version on the Microsoft official website to download.

The file directory of the folder where the online programmer software is located is shown in Figure 1-3.

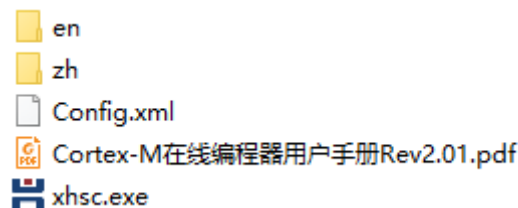


Figure 1-3 Online programmer file directory

The "en" and "zh" folders contain program language configuration files, which can realize the switching between Chinese and English in the program interface, but are not required.

"Config.xml" is the user configuration file. It is automatically generated after the program is closed to save user-related settings. This file is not available for the first use.

"Cortex-M Online Programmer User Manual Rev2.01.pdf" is the user manual file, which can be opened by clicking the "Help" menu in the program interface.

"xhsc.exe" is the program execution file. This file can also be run alone, but it cannot realize the switching between Chinese and English, and only supports English display.

Double-click "xhsc.exe" to open the software. The software interface is shown in Figure 1-4.

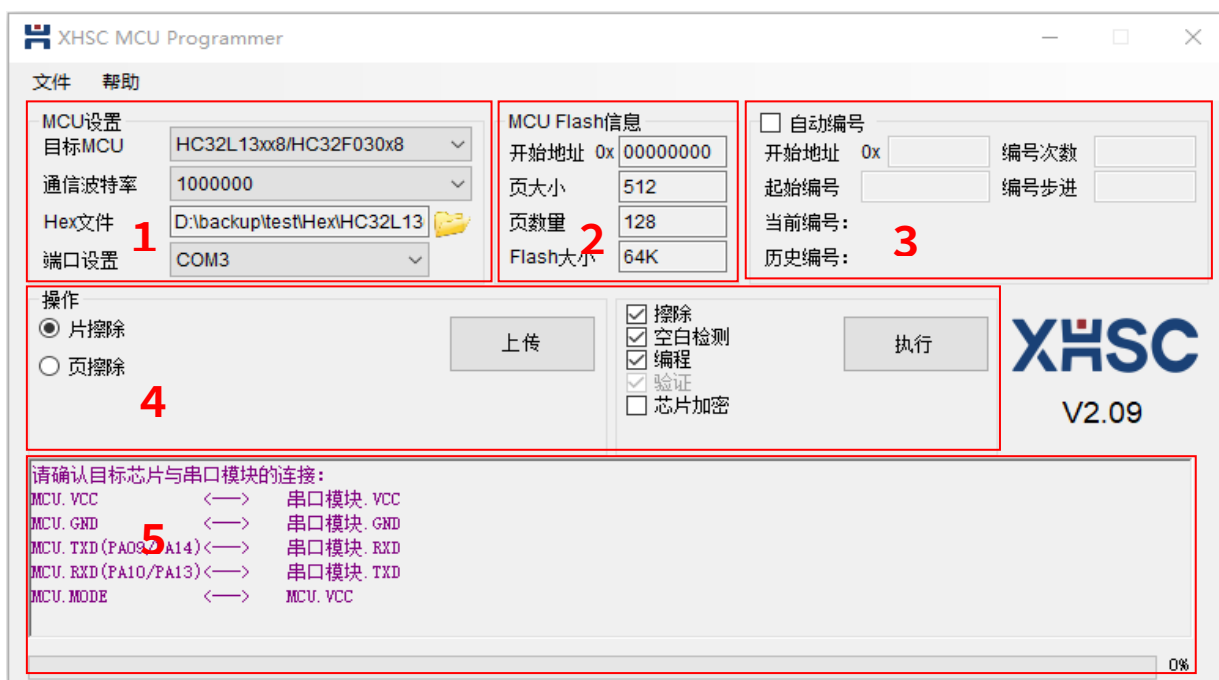


Figure 1-4 Software Interface 1

- **MCU Settings:** used to set the target MCU model, the crystal frequency used by the target MCU (HC32F146xA/ HC32M140xA and HC32F146x8/ HC32M140x8) or the communication baud rate of the selected serial port (other series), the Hex file to be burned, and the COM port number on the PC.
- **MCU Flash Information:** Displays the selected MCU Flash information: start address, page size, number of pages, and Flash size.
- **Auto Numbering:** Used for users to number the MCU.
- **Operation:** The operation is divided into two parts: upload and execution. Upload is to read the target MCU Flash data and save it as a .hex file;
Execution can select the operation item you want to operate, including erase, blank check, programming (verification), and chip encryption. After selecting the operation item, click the execute button.
- **Information Display:** Used to display programming information.

For target MCU HC32F460xExx/HC32F45xxExx, the software interface is shown in Figure 1-5.

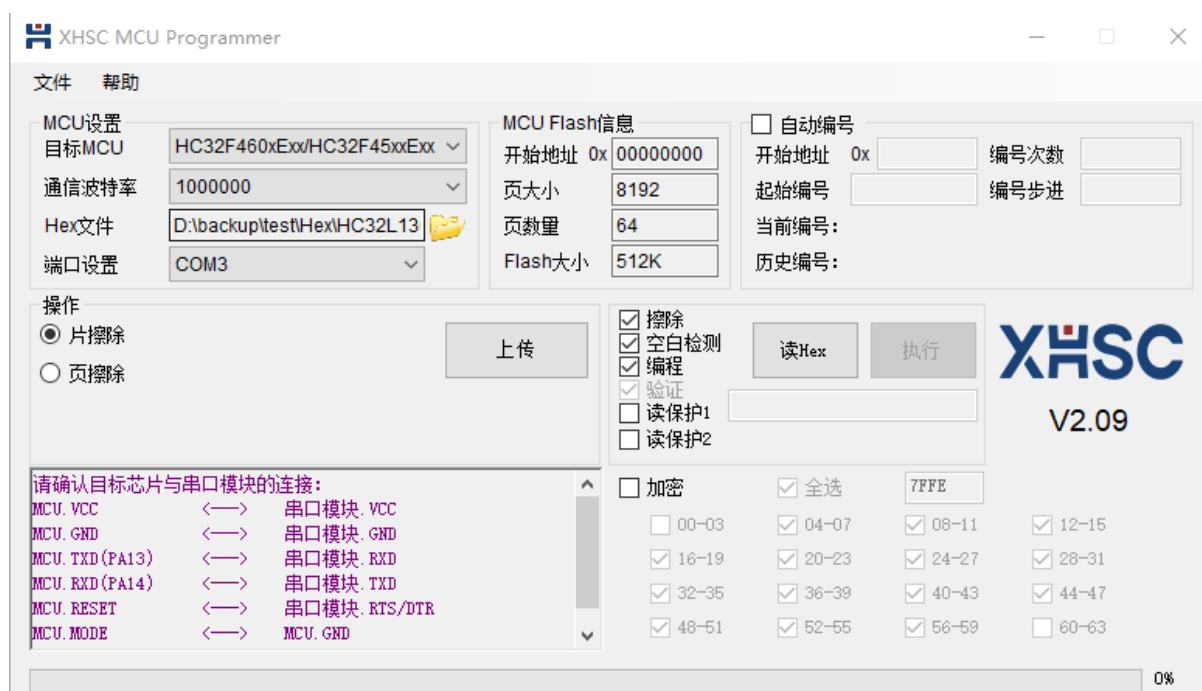


Figure 1-5 Software Interface 2

HC32F460xExx/HC32F45xxExx has read protection 1, read protection 2, and encryption functions. If you want to operate these functions, you need to select the corresponding option box. Read protection 1 requires a password, and the encryption function requires setting the flash encryption range.

2 Quick Operation

The following will describe how to quickly program.

1. Connect the serial port of the USB-to-serial cable to the serial programming pin of the target MCU. Taking the HC32L136 series as an example, the connection method is shown in Figure 2-1. Pull up the target MCU MODE pin, and then power on the target MCU to enter the serial programming mode.

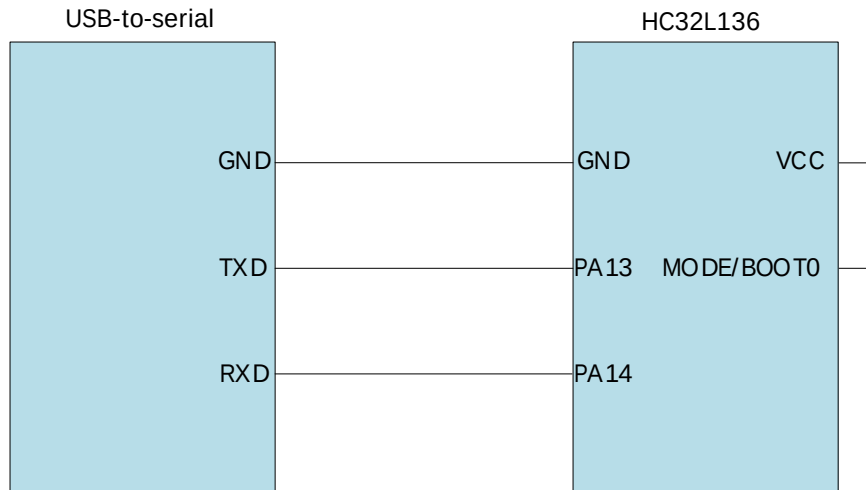


Figure 2-1 Hardware connection

2. Connect the PC and MCU target board with a USB-to-serial cable. Open the software, select the corresponding target MCU model, set the communication baud rate, select the programmed Hex file, and set the port to the COM port number used.



Figure 2-2 MCU settings

3. 3. Select the desired operation. For example, select the "Erase", "Blank Detection", "Program (Verification)" checkboxes. As shown in Figure 2-3.



Figure 2-3 Select Operation

4. Click the "Execute" button to start programming and wait for the programming to end.

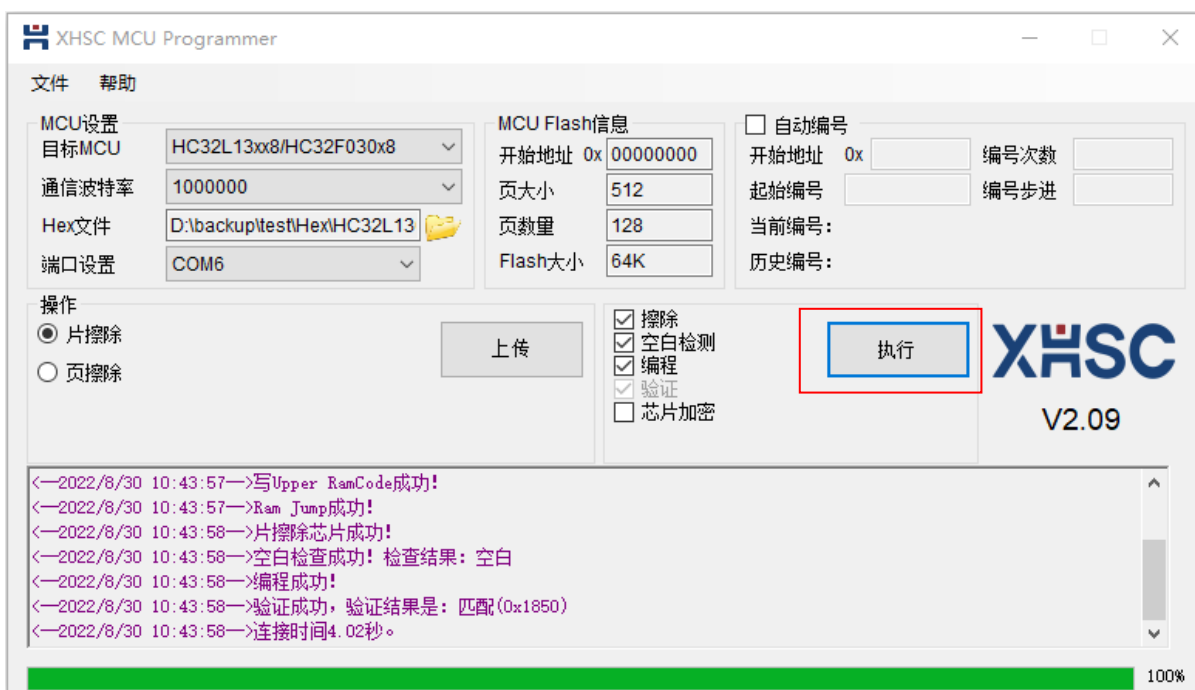


Figure 2-4 Execute Operation

3 Operation Description

3.1 MCU Settings

MCU related information settings, select the MCU model, set the crystal frequency or communication baud rate, select the Hex file to be programmed, and the corresponding port.

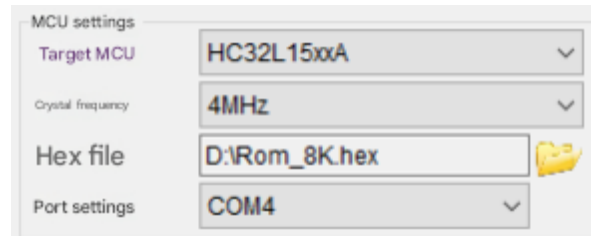


Figure 3-1 MCU Settings

- Target MCU: This drop-down box contains all the ARM cortex-M core MCU types currently available at Xiaohua. Select a model corresponding to the programming target MCU.
- Crystal frequency or communication baud rate: For HC32F146xA/ HC32M140xA and HC32F146x8/ HC32M140x8 series MCUs, this option is to set the crystal frequency used by the target MCU; for other series MCUs, this option is to set the communication baud rate of the selected port.
- Hex file: used to select the Hex file to be programmed.
- Port settings: Set the port number for ISP connection.

3.2 MCU Flash Information

The MCU Flash information of the selected model is displayed: start address, page size, number of pages and Flash size. As shown in Figure 3-2.

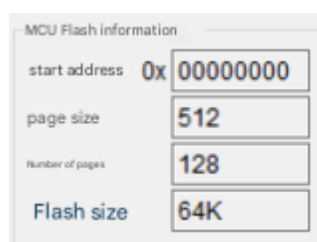


Figure 3-2 MCU Flash information

3.3 Automatic Numbering

This software has the function of numbering the target MCU when programming. You only need to select the radio button "Auto Numbering" in the "Auto Numbering" group to set the automatic numbering.

Figure 3-3 Enable automatic numbering

After selecting "Auto Numbering", the automatic numbering function is enabled, and the corresponding parameters are filled in. The relevant parameters are described as follows:

- Start address: Write the number to the MCU Flash address. The start number needs to occupy 4 bytes. The address must be within the MCU Flash range (depending on the corresponding MCU model). The address is hexadecimal, with characters from 0-F.
- Numbering count: How many numbers need to be written. The numbering times must be greater than 0. The value range is 1 to 999999.
- Start number: The start number is the starting number of the numbering, from 0 to 4, 294, 967, 295 (0xFFFFFFFF).
- Numbering step: The step between each number, the value range is 0 to 999.
- Current number: Displays the number written in this programming.
- History Number: The history number shows the number of the last successful write.

3.4 Operation

This section includes the main functions of this software, upload, erase, blank detection, programming (verification) and chip encryption.

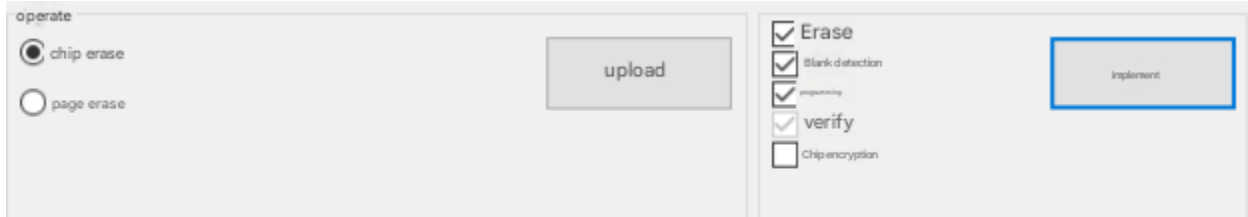


Figure 3-4 Operation

If the target MCU is selected as HC32F460xExx/ HC32F45xxExx, in addition to the above functions, it will also display: read protection 1, read protection 2 and encryption function.

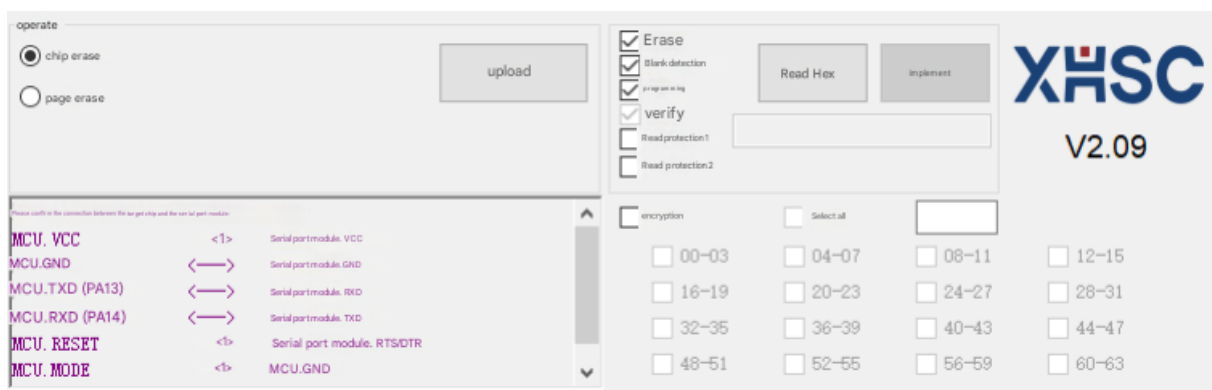


Figure 3-5 Available operations for HC32F460xExx/ HC32F45xxExx

<General operation>

- Upload: If the user has selected a file to burn, only the uploaded burned part is read;
If the user has not selected a file to burn or the burned file address is incorrect, the entire Flash content of the uploaded target MCU is read.
- Erase: Erasing includes two erase modes: chip erase and page erase. Chip erase is to erase the entire chip, and page erase is to erase according to the position of the Flash page occupied by the programmed Hex file.
- Blank detection: Check whether the entire chip is in a blank state (all 0xFF).
- Programming: Programming is the operation of programming the Hex file into the MCU.
- Verification: Verify whether the programmed code is correct.

<General protection function> (protection function of other chips except HC32F460xExx/ HC32F45xxExx)

- Chip encryption: Protect the chip flash data. After protection, the flash data cannot be read.
 - *Note: The encryption operation of HC32x07xxAxx/ HC32x17xxAxx/ HC32x19xxCxx series chips is only allowed to be performed 64 times. After 64 times, the chip is locked and the decryption operation cannot be performed.*

<HC32F460xExx/ HC32F45xxExx chip protection function>

- Read protection 1: Read protection is implemented for the FLASH area to prevent untrusted users from reading FLASH data. After protection, the data in the flash can be read through the key.
- Read protection 2: Read protection is implemented for the FLASH area to prevent untrusted users from reading FLASH data. After protection, no operation can read the data in the flash.
- Encryption: Encrypt the FLASH data to prevent the FLASH from being attacked by physical analysis. On the premise of selecting the programming option box, select the encryption option box and set the encryption sector. After the set sector is successfully programmed, the data in these sectors is encrypted.

3.5 Information Display

Used to display operation information and operation progress.



Figure 3-6 Information display

3.6 Command Line Operation

The program supports command line operations in CMD.exe or third-party programs. Take CMD.exe as an example:

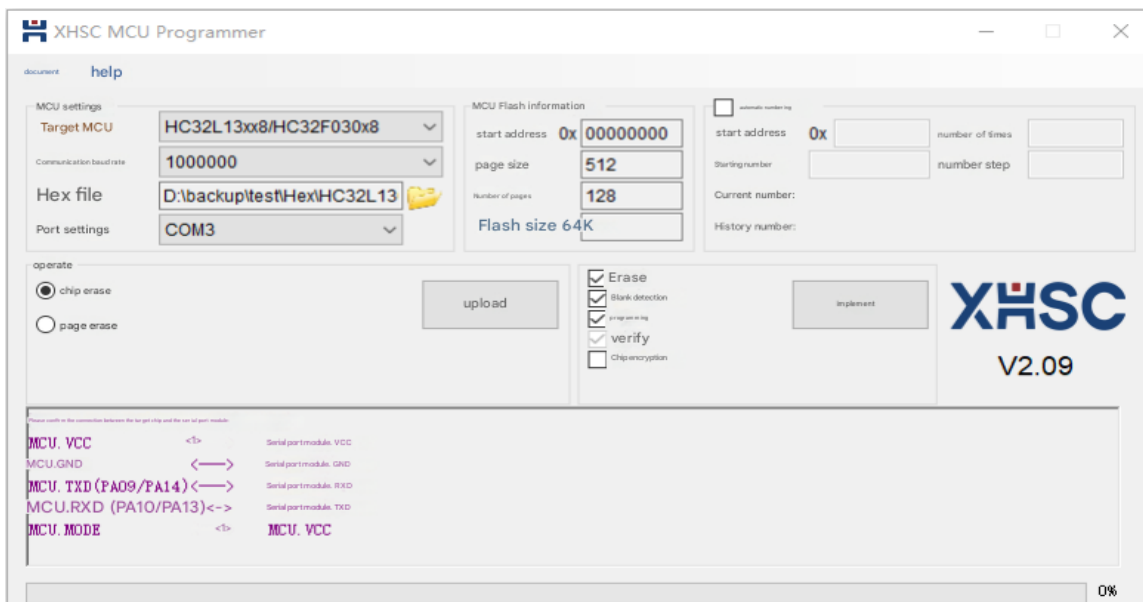
Open CMD.exe and enter the file directory where xhsc.exe is located, as shown in Figure 3-7.

```

C:\Users\xuxx>D:
D:\>cd \backup\New ISP>tag_V2.09\XHSC ISP
D:\backup\New ISP>tag_V2.09\XHSC ISP>
    
```

Figure 3-7 Enter program location directory

Type "xhsc", open the xhsc.exe program, perform "MCU settings", and then close the xhsc.exe program. As shown in Figure 3-8.



Return to CMD.exe, type xhsc ?, and the supported operation commands are displayed. As shown in Figure 3-9.

```

D:\backup\New ISP>tag_V2.09\XHSC ISP>xhsc ?
D:\backup\New ISP>tag_V2.09\XHSC ISP>
Automatic numbering-starting address-numbering times-starting number-numbering step
E: Chip erase
PE: page erase
B: Blank check
P: Programming
E: Chip encryption (read protection 1)
E2: Read protection 2 (M010)
D: Encryption (M010)
    
```

Figure 3-9 Supported operation commands

For example, enable automatic numbering, set the start address to 0x2000, set the numbering times to 1, set the starting number to 0, set the numbering step to 1, select chip erase, blank check and programming at the same time, type "xhsc AN-2000-1-0-1 CE B P" and press Enter. The operation result is shown in Figure 3-10.

```

D:\backup\New ISP\tag_V2.09\XHSC ISP>xhsc AN-2000-1-0-1 CE VE

D:\backup New ISP\tag_V2.09\XHSC ISP>Connected successfully!
Download and run RAM CODE successfully!
The board reset was changed successfully!
Supper RamCode Success!
Ram Jump successful!
Chip erased successfully!
Blank check successful! Check result: Blank
Automatic numbering: Verification successful, verification result is: Match (0x0000)
Verification successful, verification result is: match (0x1850)
Connection time 3.14 seconds.
  
```

Figure 3-10 Example operation information

Note:

- Different operation command parameters must be separated by spaces.

4 Error Handling

4.1 Tips

Table 4-1 Tips

Prompt message	Description	Processing
Connection successful	Connection successful	
The MCU has been encrypted and needs to be manually restarted. Please click "Yes (Y)" after manually restarting the MCU.	Prompt to manually restart the MCU	Manually restart the MCU, then click the "Yes" button to continue.
Automatic numbering completed	Automatic numbering completion	
The auto-numbering address overlaps with the user code. Do you want to continue?	Ask whether to continue automatic numbering	Click "Yes" to continue numbering, otherwise stop writing numbers

4.2 Errors

Table 4-2 Errors

Error message	Description	Handling
Please select the Hex file you want to burn.	No Hex file is selected.	Select Hex file.
The file path is incorrect or the file is invalid	The selected Hex file path does not exist or the file is invalid or occupied.	Reselect the Hex file and make sure the Hex file is valid and not occupied
Hex File format error	Hex File format error	Confirm whether the Hex file is correct
Hex file error! The length exceeds the selected chip Flash size.	The Hex file size exceeds the selected chip Flash size	Reselect the correct Hex file or reselect the correct target MCU
Please install a serial port on this computer	This computer has no serial port	Install a serial port
Serial port operation timeout	Communication failed	Check the hardware connection, check firmware matching, and try to power on again.
Read failed	Read failed	Check the hardware connection, check firmware matching, and try to power on again.
MCU Flash Encrypted	Flash is encrypted, data cannot be read	Encrypted chip, Flash data cannot be read out
Chip erase failed	Chip erase failed	Check the hardware connection, check firmware matching, and try to power on again.
Page erase failed	Page erase failed	Check the hardware connection, check firmware matching, and try to power on again.
Page erase failed: chip is encrypted	Encrypted chip cannot be page erased	Select Chip Erase
Blank check failed	Blank check failed	Check the hardware connection, check firmware matching, and try to power on again.
Verification and zeroing failed	Checksum reset failed	Check the hardware connection, check firmware matching, and try to power on again.
Programming failed	Programming failed	Check the hardware connection, check firmware matching, and try to power on again.
Verification failed	Verification failed	Check the hardware connection, check firmware matching, and try to power on again.
The parameter cannot be empty, please fill it in	Parameters not filled in	Fill in the parameters
The parameter format is wrong, please fill it in again	Parameter format error	Re-enter the parameters
The address exceeds the Flash size of the selected chip, please fill it again	Start address invalid	Re-enter address

Revision History

Version	Date	Description
Rev1.0	2017/11/10	First version of the Cortex-M online programmer user manual.
Rev1.1	2019/04/09	Added description of software version V1.4.
Rev1.2	2019/04/15	Added supported chip models.
Rev2.0	2019/09/03	Support for software version V2.0.
Rev2.01	2021/12/31	Models added.
Rev2.1	2022/08/30	Models added and company logo updated.
Rev2.11	2023/04/13	Models updated.
Rev2.12	2023/07/05	Removed the description of specific supported chip models.